

Resume — Tim Pečerskis, MSc

Technical entrepreneur and system algorithm designer. 15+ years turning complex system control theory into production-grade outputs — **from transistors to virtual environments**. Track record of leading cross-discipline teams, running full product lifecycles, and driving technical sales from pre-sale through delivery.

Domain Expertise

Product & Project Leadership — Product design · roadmaps for reliability & scalability · SDLC and hardware lifecycle management · agile · compliance & certification · cross-functional team leadership · hiring & org scaling.

Deep Tech — Protocol security · software engineering · electrodynamics · machine learning · blockchain · real-time signal processing · low-latency systems · data analysis · R&D · patent development and IP strategy.

Business Operations — Budget ownership · process optimisation · vendor management · stakeholder communication · compliance framework navigation.

Growth — Technical sales · presales · account management · solutions architecture · developer advocacy · public speaking · technical writing.

Professional Experience

Founder — Peeramid Labs

November 2024 — Present

Peeramid Labs is my own wrapper company for personal R&D, venture exploration, and consulting engagements. Under it I run a lean team of three, experimenting across AI infrastructure, blockchain security, and privacy — testing multiple business directions in parallel.

Skills: Team leadership · Rust · Python · TS/JS (React, Node) · LaTeX · EVM (Arbitrum, Ethereum, Reth, Celo, Optimism, zkSync, Base, smart accounts, MEV) · Circom · Foundry · wagmi · alloy-rs · Zama FHE · Web3 security · protocol engineering · CUDA · LLM inference balancing · vLLM · SGLang · pandas · scipy · sklearn · Huggingface · NATS.io · Docker · Redis · Postgres · GCP · Vercel · game-theoretic and financial modelling · P2P networking · business development · investor relations · product strategy for compliance and regulation.

Achievements:

- Led deployment of three ML products: AI code-slop firewall, bytecode-based smart contract audit tool (**83% precision, 2.4× detection lift**), and a health-prediction engine ($R^2 = 0.52$ on held-out subjects, spectral graph model over delay-embedded phase-space).
- Authored *Mixture-of-Models: Unifying Heterogeneous Agents via N-Way Self-Evaluating Deliberation* (NSED, arXiv:2601.16863) — a distributed GPU neural architecture for heterogeneous model orchestration delivering **4× CapEx reduction** or **30% accuracy boost** on any base model benchmarked.
- Wrote enterprise-grade Rust orchestration implementation of the NSED protocol — event-driven architecture on NATS as high-speed event bus, **70k LoC of code + 150k LoC of tests** including end-to-end and documentation. SDK open-sourced.
- Filed **6 patents across 5 families** — multi-agent AI coordination, novel AI architecture (zero prior art), entropy-based model reputation systems, cryptographic compliance infrastructure, and hardware-bound licensing.
- Identified and responsibly disclosed a critical vulnerability in **Ethereum EIP-7702** — full write-up.
- Built Peeramid Labs branding, company voice, and the engineering blog at blog.peeramid.xyz.

Lecturer, Blockchain Architecture & Programming — Invisible Garden Foundation, US

October 2024 — Present

Skills: Tutoring · EVM · Solidity · computer science · business strategy advisory · startup incubation ·

zero-knowledge proofs · FHE · Zama.

Achievements:

- Tutored **100+ students** over two years in blockchain fundamentals, smart-contract security, and Solidity.

Solution Architect / Senior Solution Developer — OpenZeppelin, UK

November 2022 — August 2024

Skills: Technical account management · pre-sales · solution engineering · solutions architecture · public speaking · incident response planning · audits · DevSecOps · AWS · KMS · Terraform · Cosmos SDK · security monitoring · governance monitoring · DeFi · Solidity · Rust · IaC · wargames · DataDog · Grafana.

Achievements:

- Led **technical account management and pre-sales processes** for the customer portfolio — owning discovery, scoping, technical qualification, and post-sale delivery handoff.
- Served as technical advisor on the BIS Project Mariana, contributing to the successful launch of the world's largest cross-institutional banking CBDC prototype.
- Co-authored the **EEA DeFi Risk Assessment Guidelines** — an industry-standard technical specification.
- Led automation, monitoring and incident-response service deployments on OpenZeppelin's IaC **Defender** platform, delivering near-100% risk-factor monitoring coverage for customers.
- **Zero hacks on my customer portfolio during my shift.**

Product Engineer — Moonstream (formerly Simiotics / Bugout), California, US

June 2021 — August 2022

Skills: JS · TS · React · web3.js · ethers.js · Brownie · Figma · product design · UX research · big-data analysis.

Achievements:

- Shipped two products from zero into a fast-paced web-startup environment:
 - **Bugout** — developer-tooling analytics platform.
 - **Moonstream** — build, scale and monitor a game's on-chain economy. Menu of drop-in mechanics — lootboxes, crafting recipes, achievements, leaderboards — that game developers integrate in minutes. Platform processed **\$4B+ in transaction volume** in its first months post-launch.

Head of Engineering — SIA TeknoLogic, Latvia

February 2016 — November 2020

Owned the OEM/ODM portfolio for all key accounts. Delivered **30+ distinct hardware/software products** end-to-end (concept → production) with near-100% customer retention. Customers included LeClanché, Barix, Erica Synths, Aretai Speakers, and Coastal Amusements.

Skills: C · C++ · assembly · embedded Linux · Yocto · RTOS · Qt · Altium Designer · KiCad · high-speed digital / power / mixed / analog / PCI-E system design · schematic and PCB · BoM management · ARM-M and ARM-A · Nordic nRF · STM32 · IAR · GCC · CMake · EMI/EMC certification · DFM · supply-chain development · sales · business development · resource planning.

Achievements:

- **Erica Synths** — designed the **Black Dual VCF**, which involved re-issuing the famous Curtis CEM3320 filter chip at a local Latvian semiconductor foundry as the **AS3320**. Designed the **DIY Delay full kit** — the DIY platform kit anchoring Erica Synths' developer and hacker community. **Zen Delay (with Ninja Tune)** — DSP module fully designed by me, hardware and algorithm, with clock-level optimisations saving **30% on BoM**. Received top reviews from Coldcut, The Bug, Toro y Moi and others.
- **Aretai Speakers** — owned the hardware portfolio; **Aretai Contra received the Red Dot Award in 2022**.
- **Coastal Amusements** — hands-on led the redesign of the amusement-machine motherboard to **automotive-grade standards**. **>200A load**, automotive-grade security over I²C, SPI and USB, engineered as a plug-in platform for any customer business objective.

- **LeClanché** — delivered the **Battery Management System** power circuitry for high-reliability lithium battery packs.
- **Barix** — led development and full **EMI/EMC compliance certification** for the Barix Exstreamer M400 audio-over-IP appliance.

Product Owner — Edge Autonomy (acquired by Redwire, NYSE: RDW)

March 2015 — February 2016

Managed a cross-discipline team of **15 engineers** spanning electronics and mechanical. Set up the company's internal documentation and work-guide system that unlocked reliable project delivery. Led testing of the company's long-range communication system.

Skills: Radio engineering · product ownership · documentation · training · leadership · project management · resource planning.

Achievements:

- Increased operational range of video-streaming radio links from **5 km to 140 km+** by fixing design flaws in the existing platform.
- Cleared a **2-year product backlog in a single quarter**.
- Outperformed the project roadmap by **3x**.
- Built and rolled out internal documentation and work-instruction guides that became the backbone of reliable project delivery across engineering.
- Led validation testing of the flagship long-range comms system.

Microwave / R&D Engineer — SAF Tehnika

June 2011 — February 2015

Core contributor to the company's flagship **Integra** microwave radio, including microwave filter and orthomode transducer (OMT) design, plus R&D validation for advanced concepts such as **orbital angular momentum (OAM) antennas**.

Skills: Microwave engineering (5.6–60 GHz) · low-noise amplifiers · passive circuits · microstrip design · CST Studio · Matlab · RF synthesis (Genesys) · Altium Designer · documentation · LoRa · DFM.

Achievements:

- Designed and patented the core signal-processing algorithm for the **Spectrum Compact** — a world-first hand-held battery-powered spectrum analyzer that became SAF Tehnika's best-earning product.
- Core contributor to the **Integra** microwave radio backbone: microwave filters, orthomode transducer (OMT) design, and R&D validation for orbital angular momentum antenna concepts.
- Designed microwave circuits for flagship products — ultra-wide-bandwidth **3–6 GHz amplifiers and LNAs**, mixers, polarisation splitters, microwave filters, waveguides, antennas. Typically outperformed timelines by **~2x**.
- Cut manufacturing costs by **30%+** via Monte-Carlo analysis of manufacturing tolerances.
- Supported **Aranet** sensors as product engineer — mechanical waterproof 3D-print design, RF BoM review supporting **10 km+ range with 1-year+ battery life**.

Patents & Publications

- **Mixture-of-Models: Unifying Heterogeneous Agents via N-Way Self-Evaluating Deliberation.** 2026. doi.org/10.5281/zenodo.18338777
- **EEA DeFi Risk Assessment Guidelines — Version 1.** EEA Technical Specification, 17 July 2024.
- **Spectrum Analyzer of Ultra-Low Intermediate Frequency.** LV 14855A, 2014, SAF Tehnika.
- **6 patents filed across 5 families** — multi-agent AI coordination · novel AI architecture (zero prior art) · entropy-based model reputation · cryptographic compliance infrastructure · hardware-bound licensing.

Education

- **MSc, Electrical and Electronics Engineering** — Riga Technical University.
- **BSc, Electrical and Electronics Engineering** — Riga Technical University.

Languages

- English — Full Professional
- Latvian — Bilingual
- Russian — Bilingual
- Spanish — Limited
- Chinese — Elementary (studying)

After Work

Watersports (surfing, diving, freediving) · cycling · yoga · hiking · climbing · music (production, SoundCloud) · woodworking · retro / MMORPG gaming (retired WoW / L2 raid master).